

AI in Genomics 2026

Tunisia · 14–18 Sept 2026

Practical 4: Design

Presenter: Yu Sugihara

PROGRAMME



01 Predict



02 Interpret



03 Compare



04 **Design**



On its own, a genome sequence is just a string of letters

Genome sequence

```
TAAAGCTCGAGGTTGATAAACAAGCGTGCCGCAAAT  
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT  
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG  
AACCAGCACCAACTCOGGCAGGCTTATGCCCTGTGC  
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

FASTA

Finding the genes in it used to need extra evidence

Genome sequence

```
TAAAGCTCGAGGTTGATAACAAGCGTGCCGCAAAT  
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT  
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG  
AACCAGCACCAACTCOGGCAGGCTTATGCCCTGTGC  
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

FASTA

Where are the genes?

RNA-seq, homologues,
manual curation

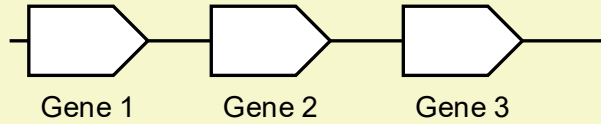
hard without extra evidence

AI now annotates a genome from the sequence alone

Genome sequence

```
TAAAGCTCGAGGTTGATAAACAAGCGTGCCGCAAAT  
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT  
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG  
AACCAGCACCAACTCOGGCAGGCTTATGCCCTGTGC  
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

Genome annotation



FASTA

AI Genome Annotation
(e.g. Helixer)

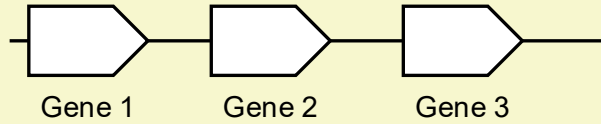
GFF

Genome annotation gives protein sequences

Genome sequence

```
TAAAGCTCGAGGTTGATAAACAAGCGTGCCGCAAAAT  
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT  
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG  
AACCAGCACCAACTCOGGCAGGCTTATGCCCTGTGC  
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

Genome annotation



Protein sequences



Protein domain annotations

FASTA

AI Genome Annotation
(e.g. Helixer)

GFF

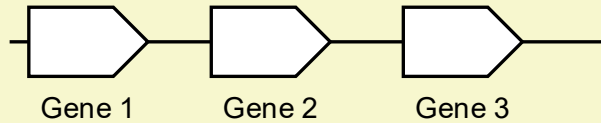
FASTA

Sequence still tells you a lot: domains and families

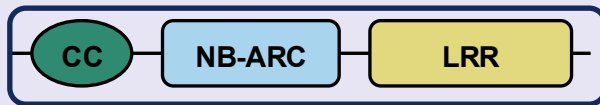
Genome sequence

```
TAAAGCTCGAGGTTGATAAACAAGCGTGCCGCAAAAT  
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT  
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG  
AACCAGCACCAACTCOGGCAGGCTTATGCCCTGTGC  
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

Genome annotation



Protein sequences



Protein domain annotations

FASTA

AI Genome Annotation
(e.g. Helixer)

GFF

FASTA

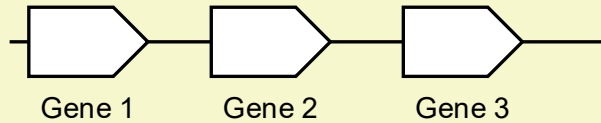
**Practical 3 · Exercise 1 — sequence
alone stops here**

But sequence cannot tell you how the domains fold together

Genome sequence

```
TAAAGCTCGAGGTTGATAACAAGCGTGCCGCAAAAT
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG
AACCAGCACCAACTCOGGCAGGCTTATGCCCTGTGC
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

Genome annotation



Protein sequences



Protein domain annotations

not possible yet

FASTA

AI Genome Annotation
(e.g. Helixer)

GFF

FASTA

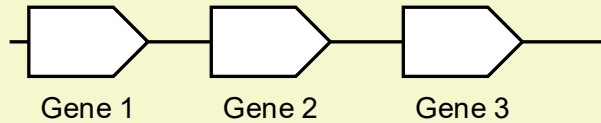
?
how do the domains fold
together?

AI structure prediction solved this

Genome sequence

```
TAAAGCTCGAGGTTGATAACAAGCGTGCCGCAAAAT  
CGGAAGCATCAAACGTATTCTACCAGGTTCTTAGT  
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG  
AACCAGCACCAACTCGGCAGGCTTATGCCCTGTGC  
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

Genome annotation

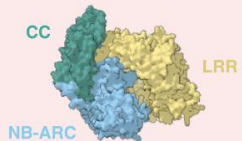


Protein sequences



Protein domain annotations

3D protein structures



FASTA

AI Genome Annotation
(e.g. Helixer)

GFF

FASTA

**AI Protein Structural
Modelling**
(e.g. AlphaFold)

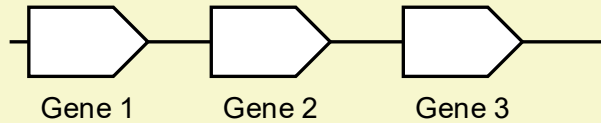
PDB

A structure can be compared even when the sequence cannot

Genome sequence

```
TAAAGCTCGAGGTTGATAACAAGCGTGCCGCAAAAT  
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT  
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG  
AACCAGCACCAACTCOGGCAGGCTTATGCCCTGTGC  
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

Genome annotation

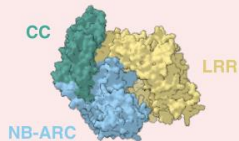


Protein sequences



Protein domain annotations

3D protein structures



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(e.g. Helixer)

GFF

FASTA

**AI Protein Structural
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PDB

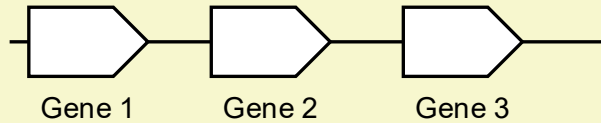
**Practical 3 · Exercises 2–4 — no domain
annotation? compare the fold instead**

The same models predict how proteins interact

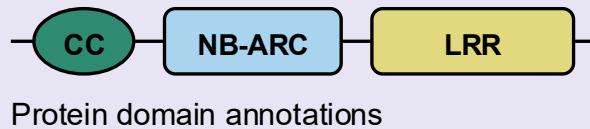
Genome sequence

```
TAAAGCTCGAGGTTGATAAACAAGCGTGCCGCAAAAT  
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT  
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG  
AACCAGCACCAACTCGGCGAGGCTTATGCCCTGTGC  
TTTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

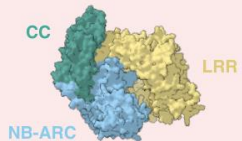
Genome annotation



Protein sequences



3D protein structures



FASTA

AI Genome Annotation
(e.g. Helixer)

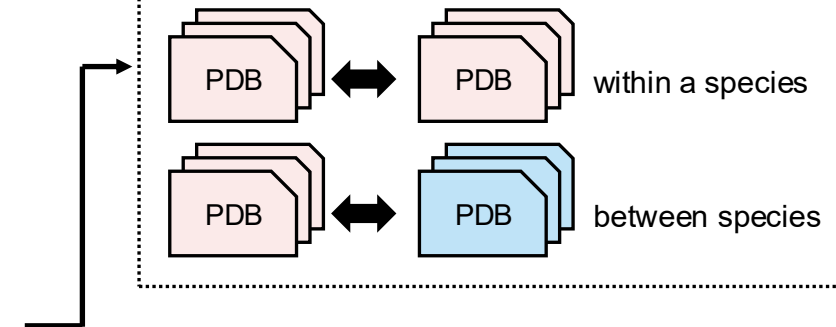
GFF

FASTA

AI Protein Structural Modelling
(e.g. AlphaFold)

PDB

Protein-protein interaction modelling

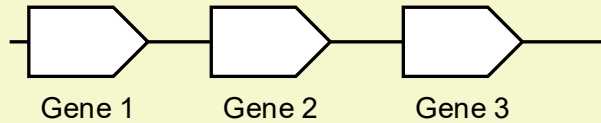


If you know the shape, you can design a new protein to bind it

Genome sequence

```
TAAAGCTCGAGGTTGATAAACAAGCGTGCCGCAAAAT
CGGAAGCATCAAACGTATTCTACCAGGCTTCTTAGT
TGTATAAAGCGCGCTCTGGTCGGCATCTGCCAATGG
AACCAGCACCAACTCGGCAGGCTTATGCCCTGTGC
TTCAAATTTTCATGTTCAAGACTCTATACACCCCG
```

Genome annotation

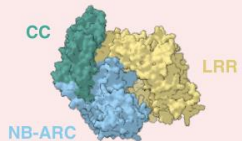


Protein sequences



Protein domain annotations

3D protein structures



FASTA

AI Genome Annotation
(e.g. Helixer)

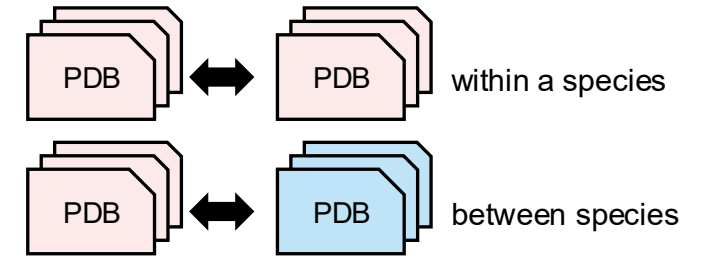
GFF

FASTA

AI Protein Structural Modelling
(e.g. AlphaFold)

PDB

Protein-protein interaction modelling



De novo binder design

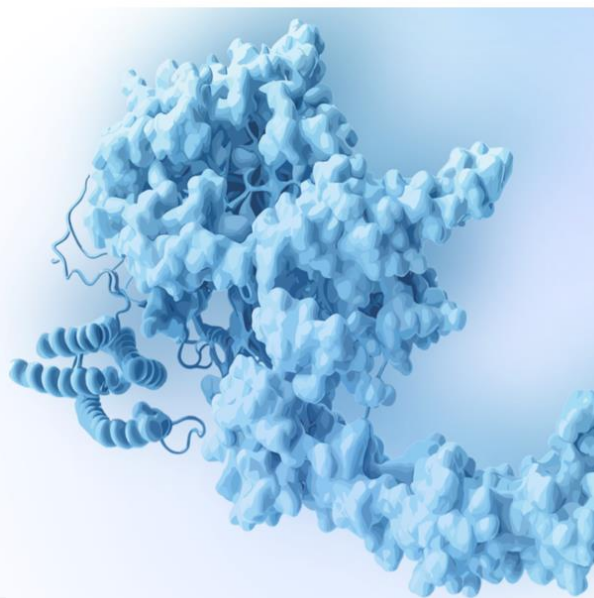
A predicted structure is a target. Practical 4.

Adaptyv Bio runs open binder-design competitions

December 25, 2024 insight

Protein Design Competition

Has binder design been solved?



Proteinbase

Proteins Collections Design Methods Targets Competitions Publish Download

Competitions

Search by title or organiser... Sort: Start Date

Ended - Results available

GEM x Adaptyv: RBX1 Binder Design Competition

RBX1 300 proteins validated

Design protein binders targeting RBX1, a cancer-relevant E3 ubiquitin ligase component. Co-organized with the GEM Workshop at ICLR 2026.

by GEM Workshop , Adaptyv Bio

Ended - Results available

Nipah Binder Competition

Nipah Virus Glycoprotein G 1200 proteins validated

Design a protein capable of neutralizing the Nipah virus, a pathogen with up to 75% mortality rate and high pandemic potential, currently lacking effective treatments.

by Adaptyv Bio

Ended

Adaptyv EGFR Binder Design Competition 2

EGFR 400 proteins validated

Design novel protein binders targeting and neutralizing EGFR, a cell-surface receptor that promotes uncontrolled cell growth in many cancers.

by Adaptyv Bio

Ended

Adaptyv EGFR Binder Design Competition 1

EGFR 200 proteins validated

Design novel protein binders targeting EGFR, a cell-surface receptor that promotes uncontrolled cell growth in many cancers.

by Adaptyv Bio

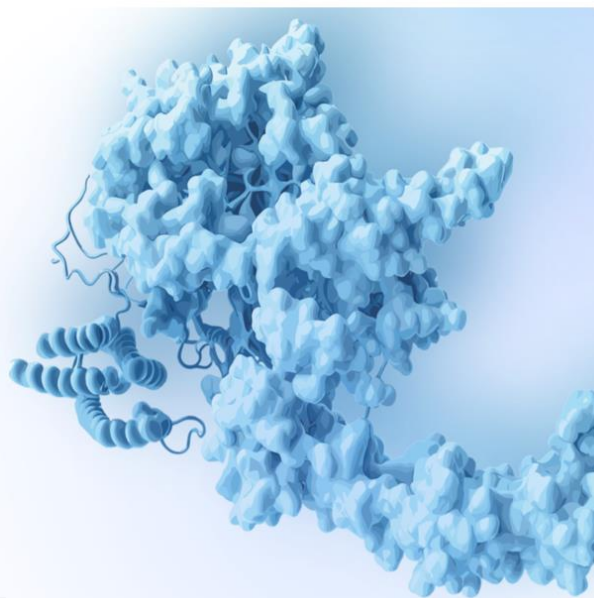
<https://proteinbase.com/competitions>

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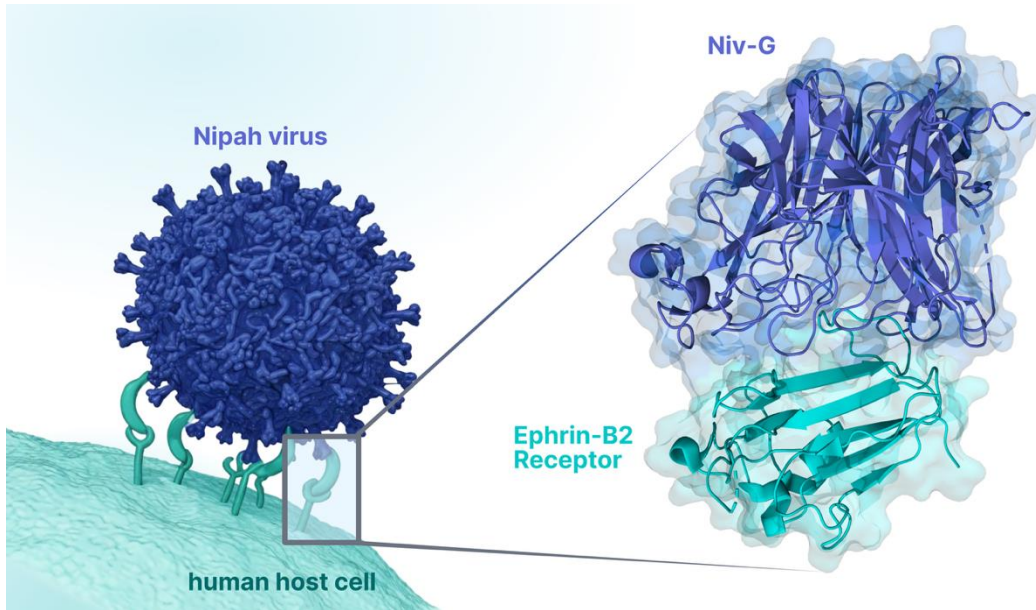
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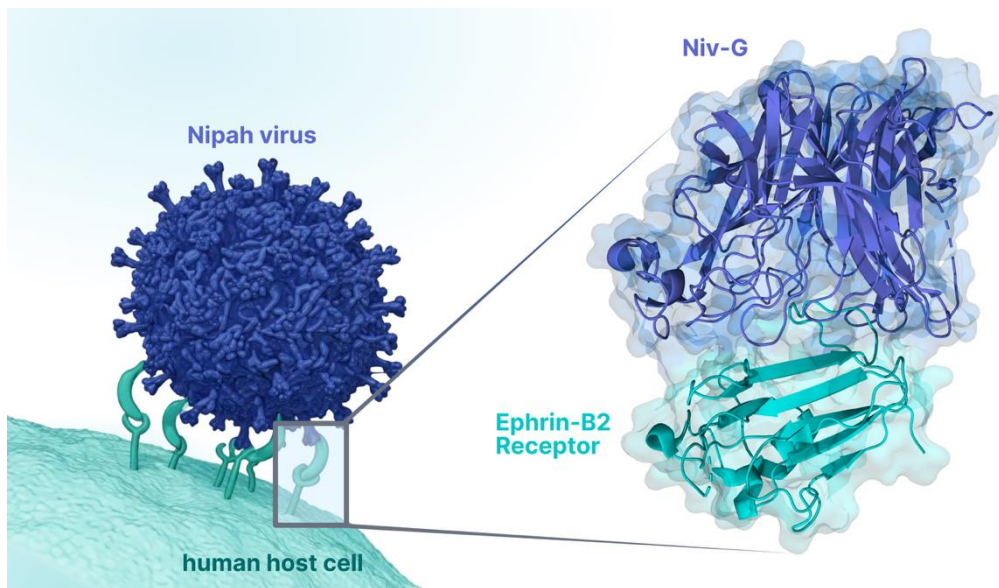
Why is Nipah virus a global health concern?



- Fatality rate of 45–70%
- Severe disease, human-to-human transmission
- Recurring outbreaks
- No approved treatment or widely available vaccine

What happened in the Nipah Protein Design Competition so far?

Designing binders against viruses



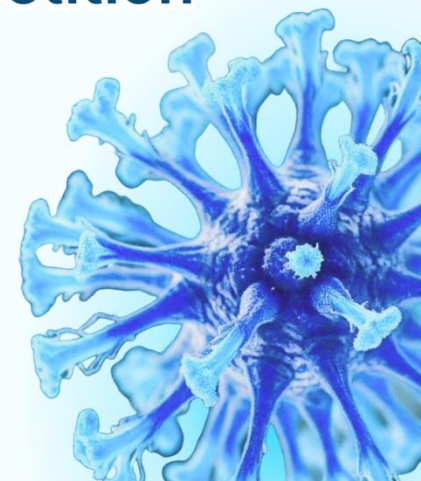
Protein Design Competition

3

680 protein designers

10'000 novel designs submitted

1'200 proteins will be validated experimentally in the wet lab



How were candidate binders selected for experimental validation?

Selection route	Designs selected	Rationale
Boltz-2 ipSAE ranking	600	Top-ranked designs based on computational interface-confidence scoring
Expert selection	400	Designs manually selected by protein-design experts
Community selection	200	Designs selected through community voting
Total	1,200	Advanced to experimental validation

**99 successful binders from 1,200
designs tested — a hit rate of about 8%**

The results of the Nipah Protein Design Competition are out!

✚ 1200 proteins experimentally validated (3x more than last year)

📈 99 novel binders against the target protein (a challenging tetramer with little prior work)

💪 26 single digit nM or better binders, with the best ones at single-digit picomolar affinity!

All data now available open-source on Proteinbase!

Let's take a look at the results 📄



6:08 AM · Jan 22, 2026 · 87.5K Views

5

43

142

60

↑

X @adaptyvbio

Nipah Binder Competition

Task: Protein Binding Target: Nipah Virus Glycoprotein G 1200 proteins experimentally validated

Design a protein capable of neutralizing the Nipah virus, a pathogen with up to 75% mortality rate and high pandemic potential, currently lacking effective treatments.

Organised by: Adaptyv Bio

Proteins (1201)

Search by name, author, or design method...

Sort: KD

Filter

Table

Grid

Miles McGibbon

#1

id: scarlet-raven-maple

Binder Other



Target Nipah Virus Binding Strong KD 3.7e-10 M

Liviu Copoiu

#2

id: lunar-mole-fern

Binder Other



Target Nipah Virus Binding Strong KD 4.5e-10 M

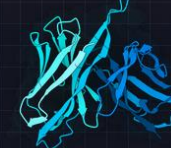
Andrew Huang

#3

id: rough-bison-maple

Binder scFv

ESMfold, RFdiffusion, and rational design



Target Nipah Virus Binding Strong KD 7.2e-10 M

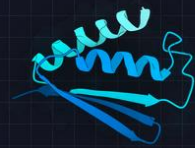
simsjn

#4

id: golden-swan-wave

Binder MiniProtein

RFdiffusion



Adaptyv Bio

#5

id: deep-heron-rose

Binder Other

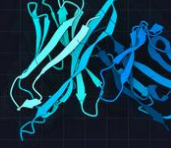


Jude Wells

#5

id: wild-yak-marble

Binder scFv



Nanobody 13% Peptide 8%
ScFv 4% Affibody <1%
Unknown <1%

Binding Strength

None 90% Strong 6%
Medium 4% Weak <1%

Expressed

True 86% False 14%

De Novo

Yes 53% No 32%
Unknown 14%

Classification

Mainly Alpha 47% Alpha Beta 26%
Mainly Beta 23% Unknown 4%
Few Secondary Structures <1%

Architecture

Unknown 80% Sandwich 10%
Up Down Bundle 4%
Orthogonal Bundle 2% Roll 1%
Alpha Horseshoe <1%
2 Layer Sandwich <1%

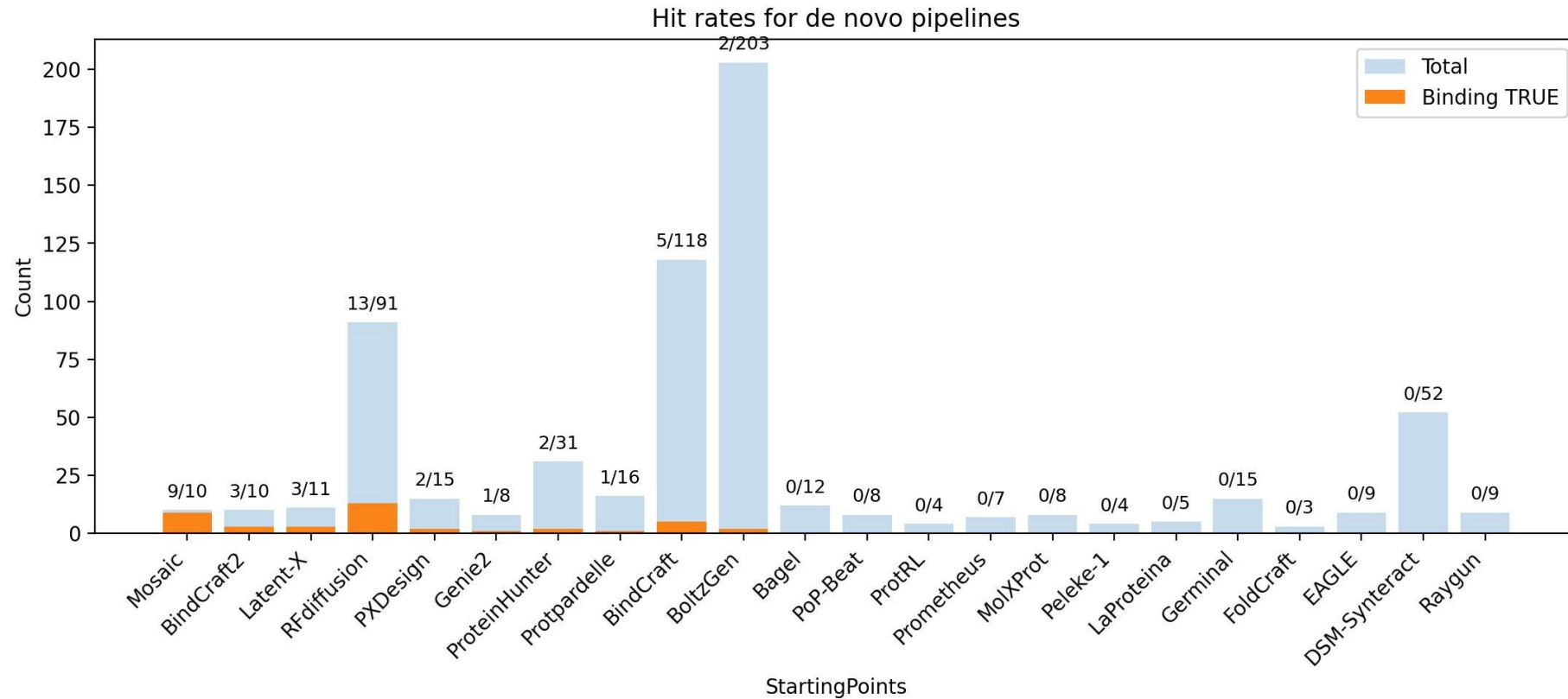
Hit rates differ between tools, but all of them are low



Martin Pacesa
@MartinPacesa

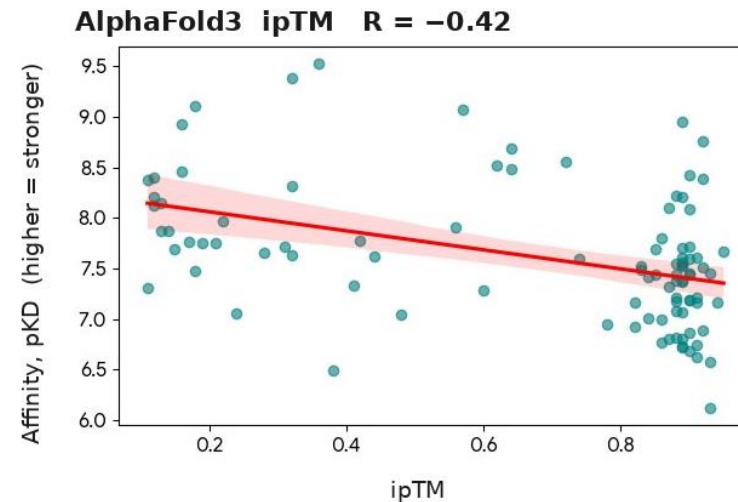
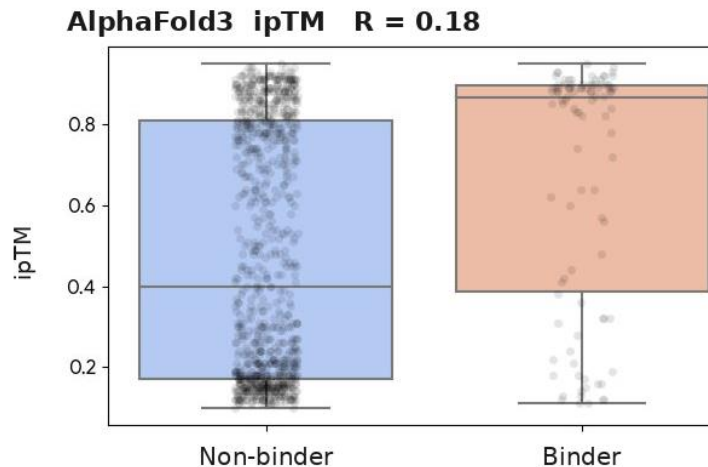
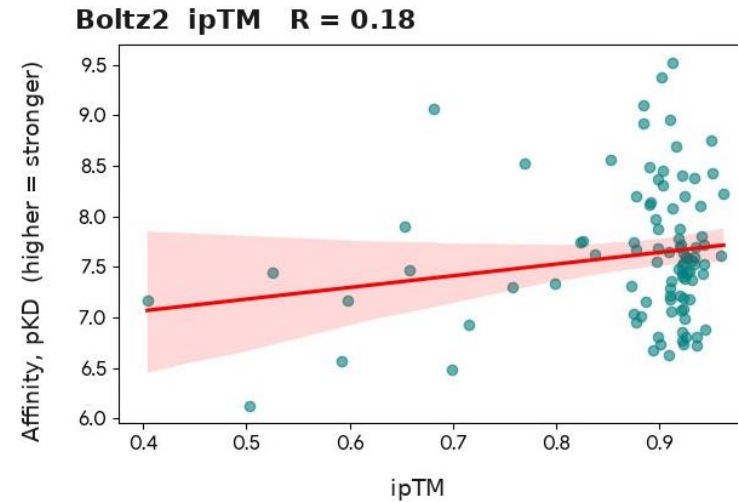
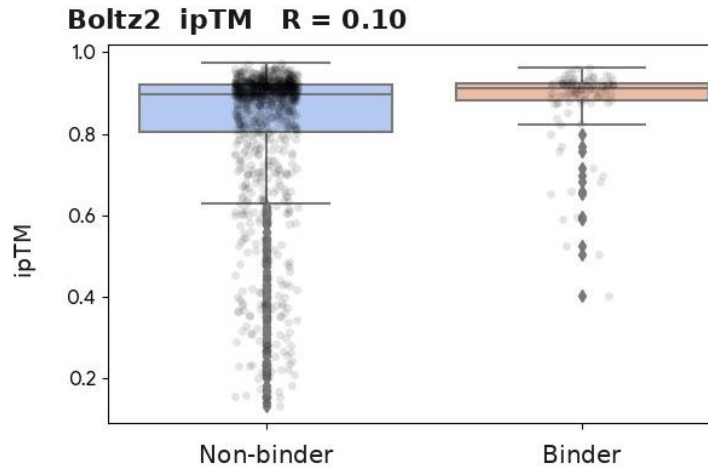


Here are the success rates of de novo pipelines based on which designs I could actually identify the methods for.



Computational scores do not predict affinity

Only ipTM is shown here. ipSAE gives the same picture.



Brian Naughton

@btnaughton

Combining data from proteinbase.com/collections/nipah/ and yehlincho.github.io/Adaptiv_Nipah_...

(w Gemini and Claude to make sure they agreed)

- AF3 predicted binding better than Boltz2
- ipSAE a bit better than ipTM
- nothing really predicted Kd (sensitive to outliers)
- see

Analysis by Brian Naughton (@btnaughton), on data from the Adaptiv Nipah competition.

x.com/btnaughton/status/2014762908048359654

Experimental hit rates vary widely across binder-design challenges

Competition	Target	Submission period	Proteins tested	Binders	Experimental hit rate
EGFR Round 1	EGFR	Jul 18 – Aug 18, 2024	201	5	2.5%
EGFR Round 2	EGFR	Oct 4 – Nov 4, 2024	400	53	13.3%
Nipah Binder Competition	Nipah virus glycoprotein G	Oct 30 – Dec 1, 2025	1,196	111	9.3%
RBX1 Binder Competition	RBX1	Feb 27 – Mar 26, 2026	321	9	2.8%
TREM2 Hackathon	TREM2	Feb 28, 2026 <i>(one-day hackathon)</i>	100	37	37.0%

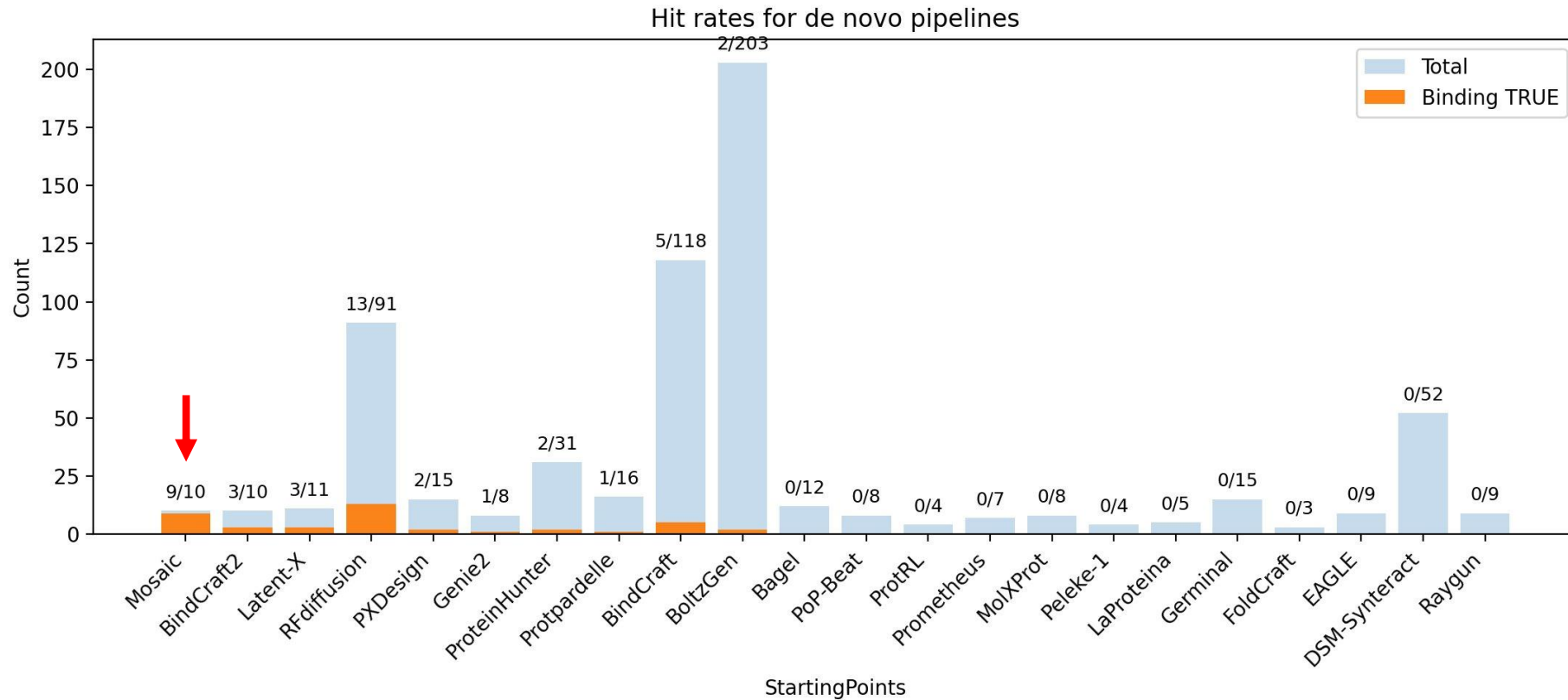
Mosaic powered the winning designs in the Nipah binder competition



Martin Pacesa
@MartinPacesa



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ESCALANTE

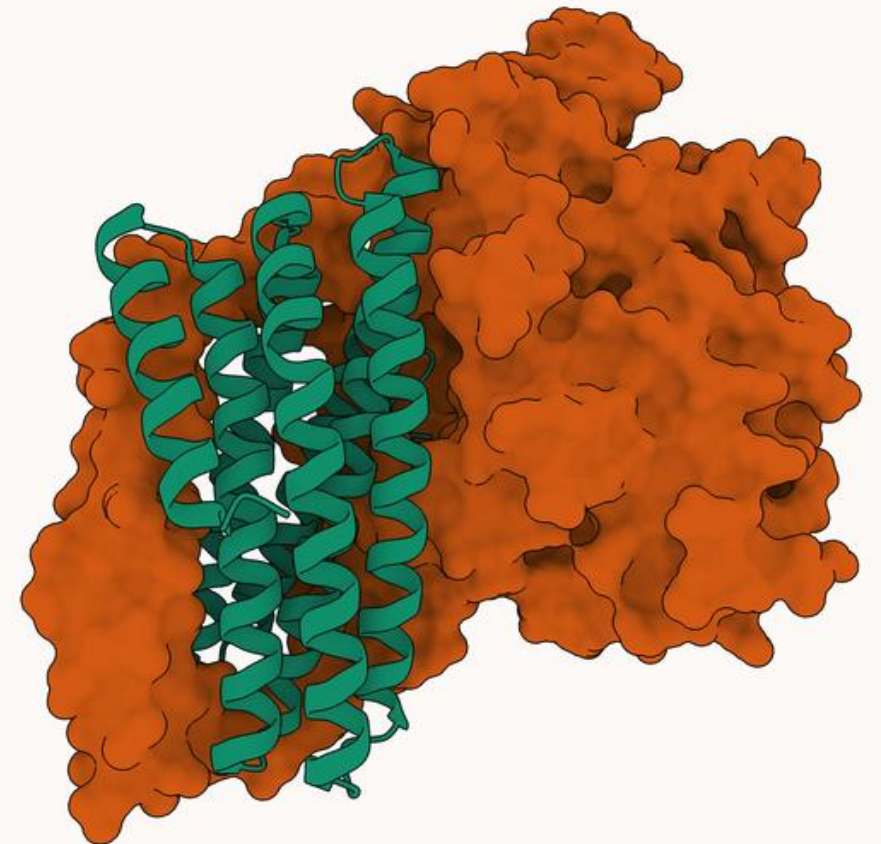
Winning the de novo portion of the Adaptyv Nipah binder competition

Nick Boyd, Sam Guns - 29 Jan, 2026

**To keep up with binder design
you have to read beyond journals**

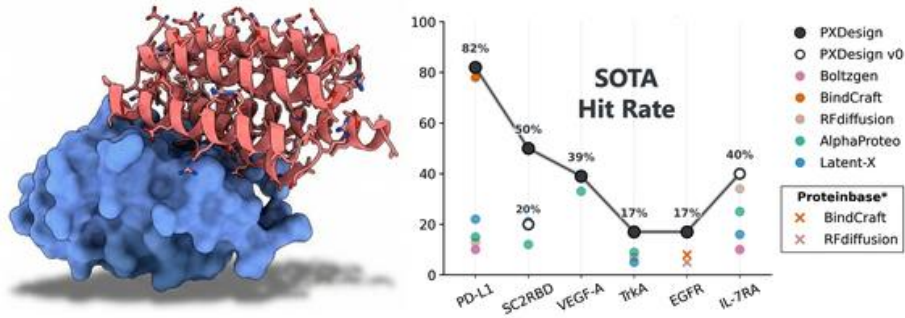
Published only as a blog post!

<https://blog.escalante.bio/winning-the-de-novo-portion-of-the-adaptyv-nipah-binder-competition/>

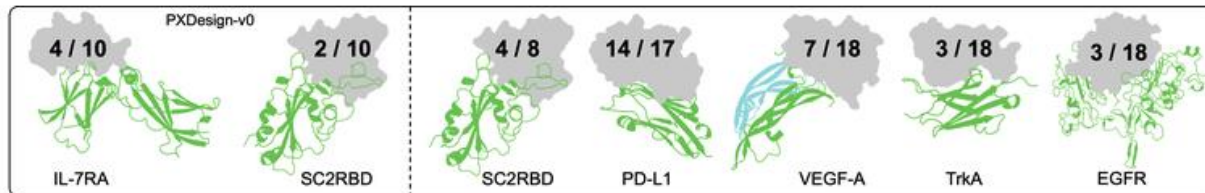


PXDesign

Fast, Modular, and Accurate *De Novo* Design of Protein Binders



<https://protenix-server.com/login>



Register with your email address — it takes a minute.

[Project Page](#) [Technical Report](#) [Web Server](#)

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Protenix Server

Protenix is an advanced AI tool for fast, accurate, and user-friendly predictions of 3D structures of all kinds of molecules, including proteins, ligands, ions, DNA, RNA, and covalent modifications.

Powered by Protenix for structure prediction, **PXDesign** is our model suite for accurate de novo binder design [\[Learn more on the PXDesign page →\]](#).

The Protenix server is free for all applications, including commercial use. We also offer our [training and inference code](#).

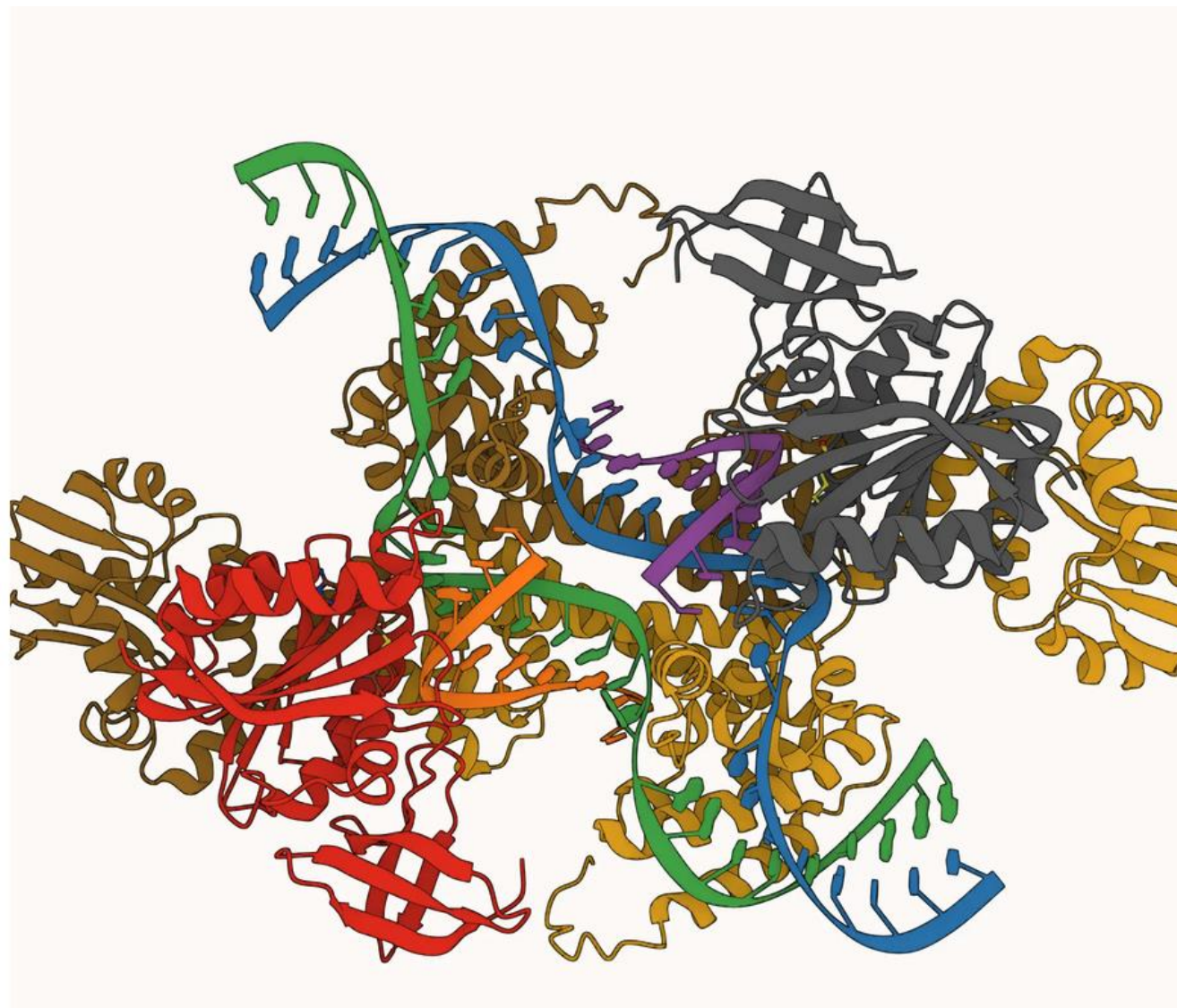
Sign in

We'll email you a magic link.

If you couldn't receive the email, please check your spam folder or add bytedance.com as a trusted sender.

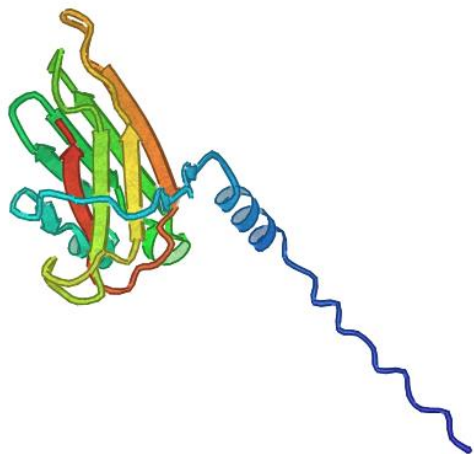
Email

Sign in

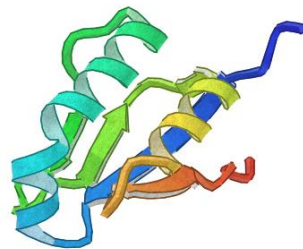


Fusarium oxysporum f. sp. *albedinis* causes dieback of date palm

FoaToxA



FoaKP6

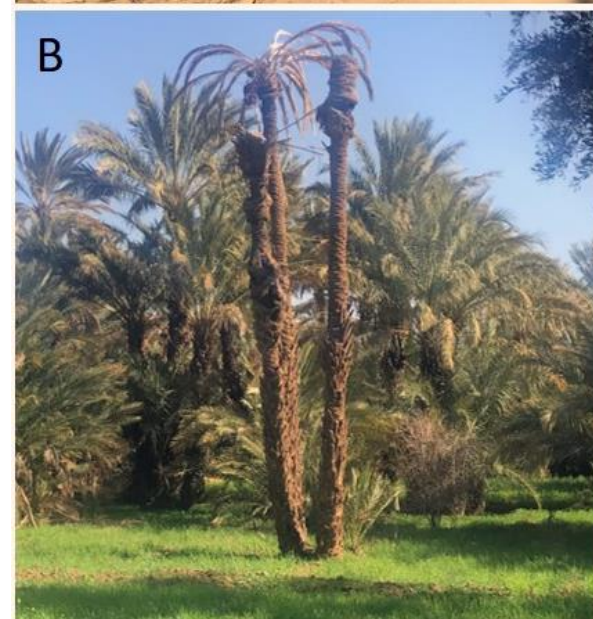


Dieback of date palm in a natural oasis in Morocco
Rafiqi et al., 2022 · PMID 35617325

A



B

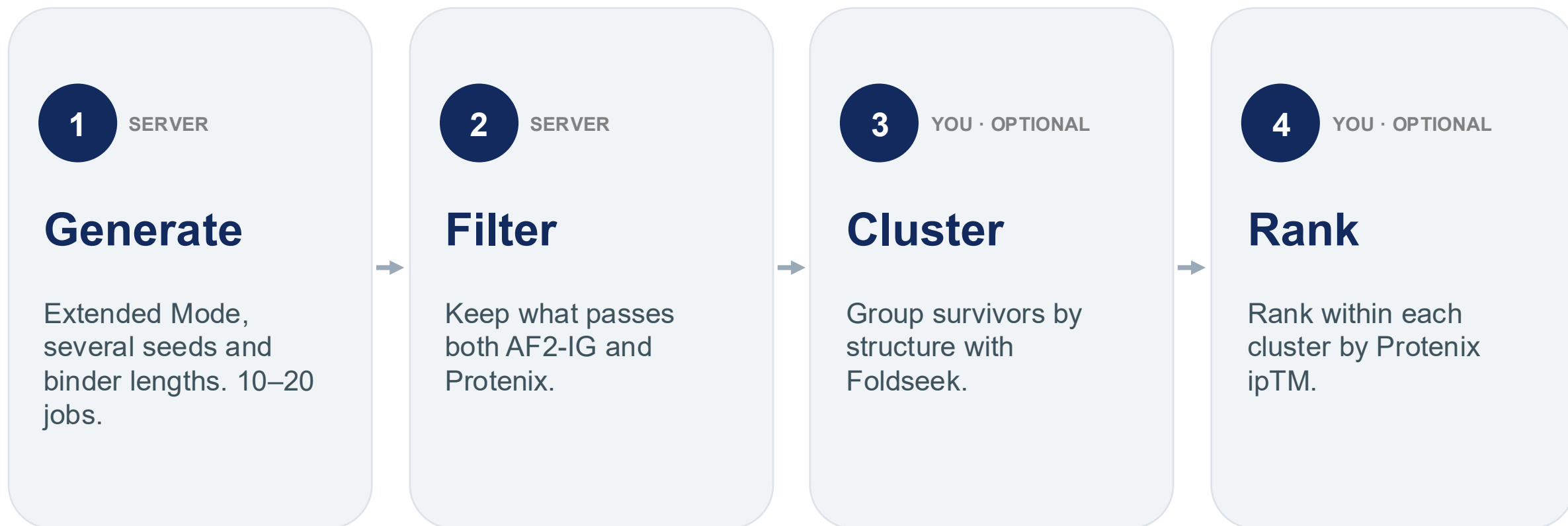


C



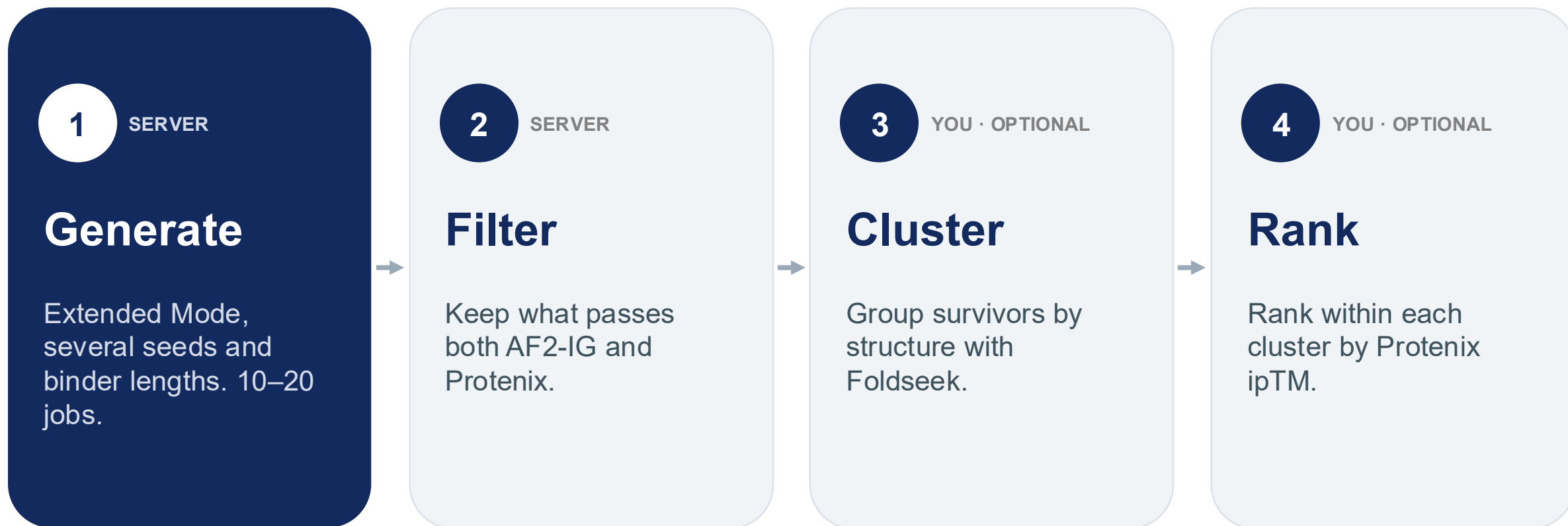
PXDesign's own recommendation for choosing designs

PXDesign's own advice, from protenix-server.com. Not a general rule.

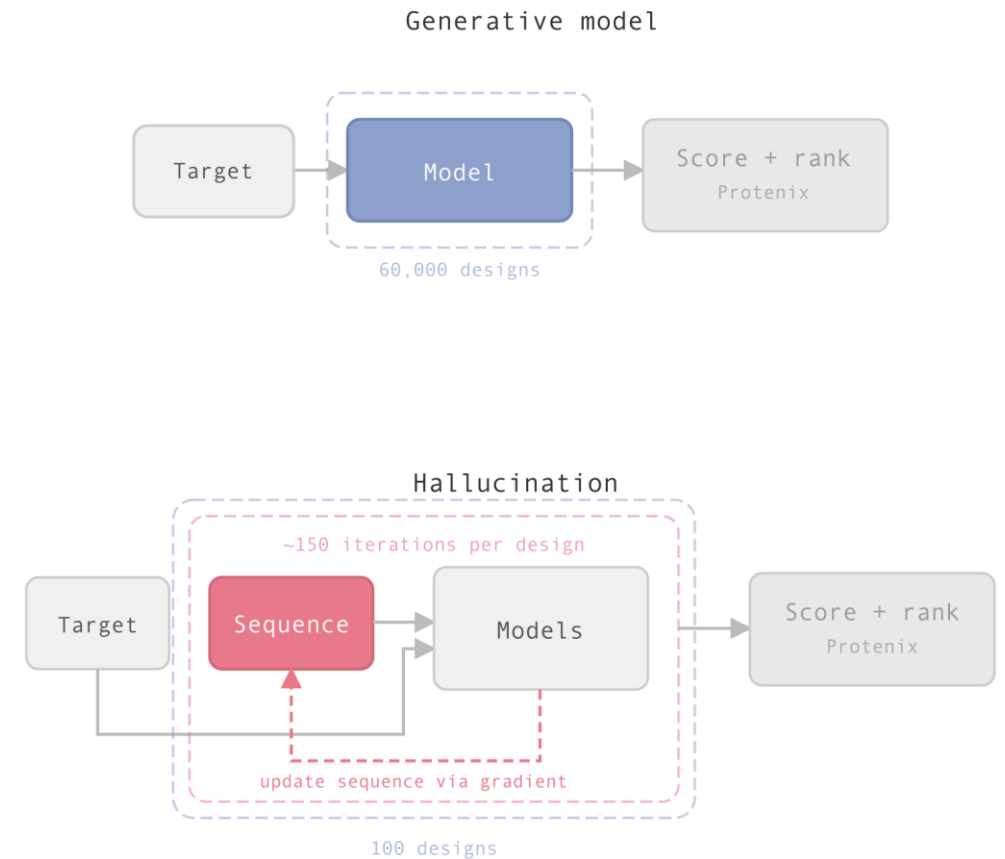
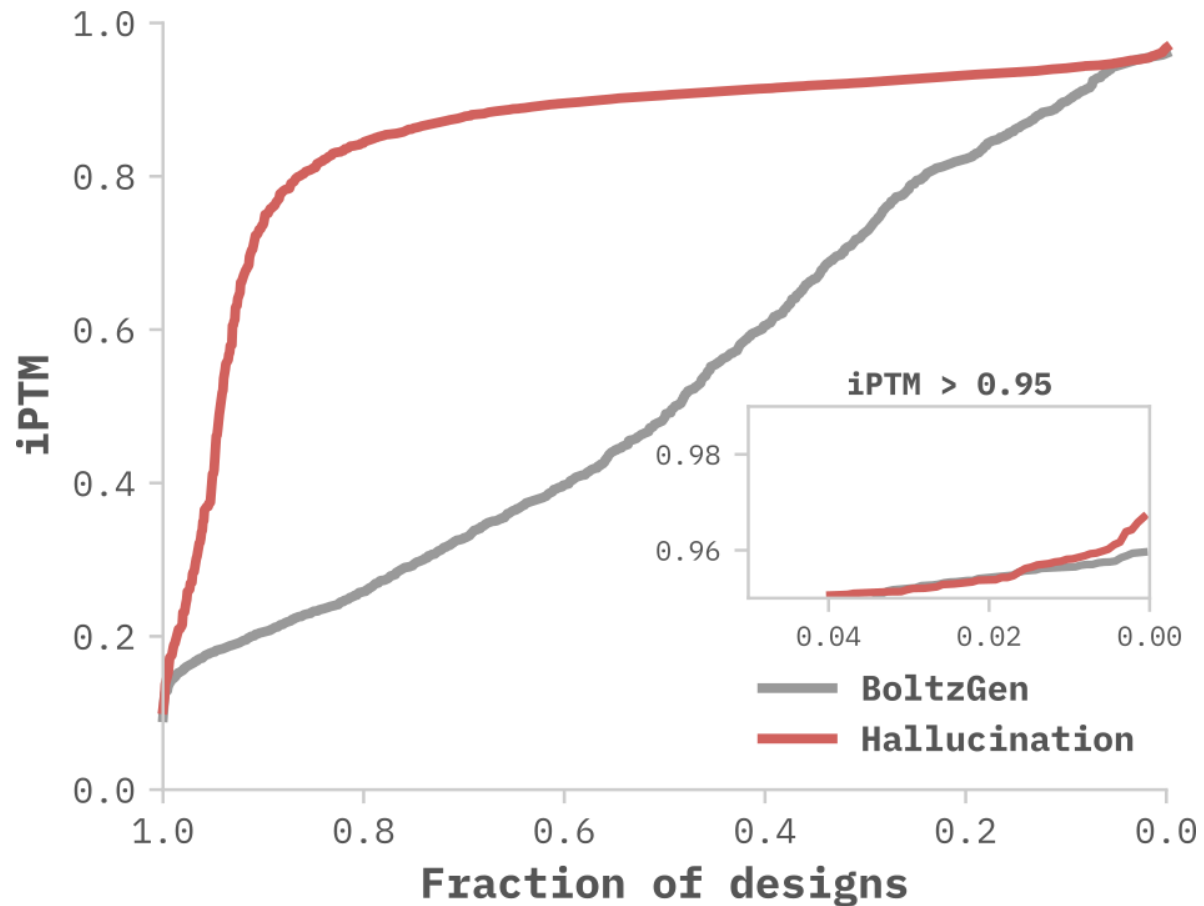


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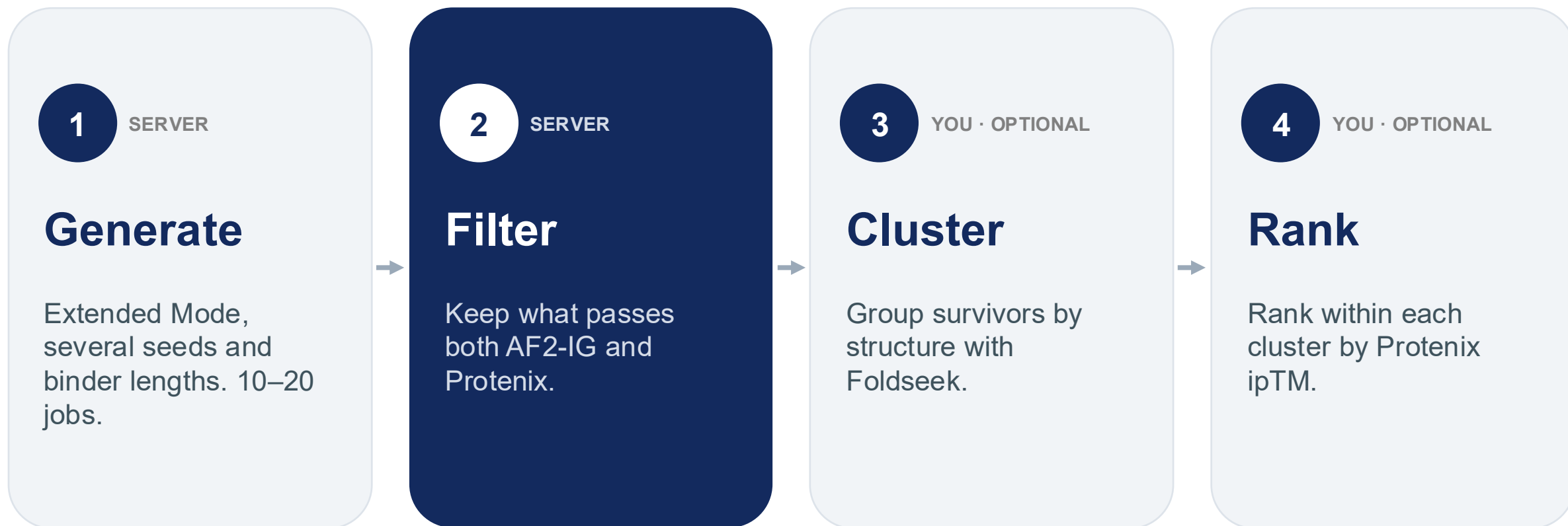


Diffusion is faster; hallucination gives better designs per sample



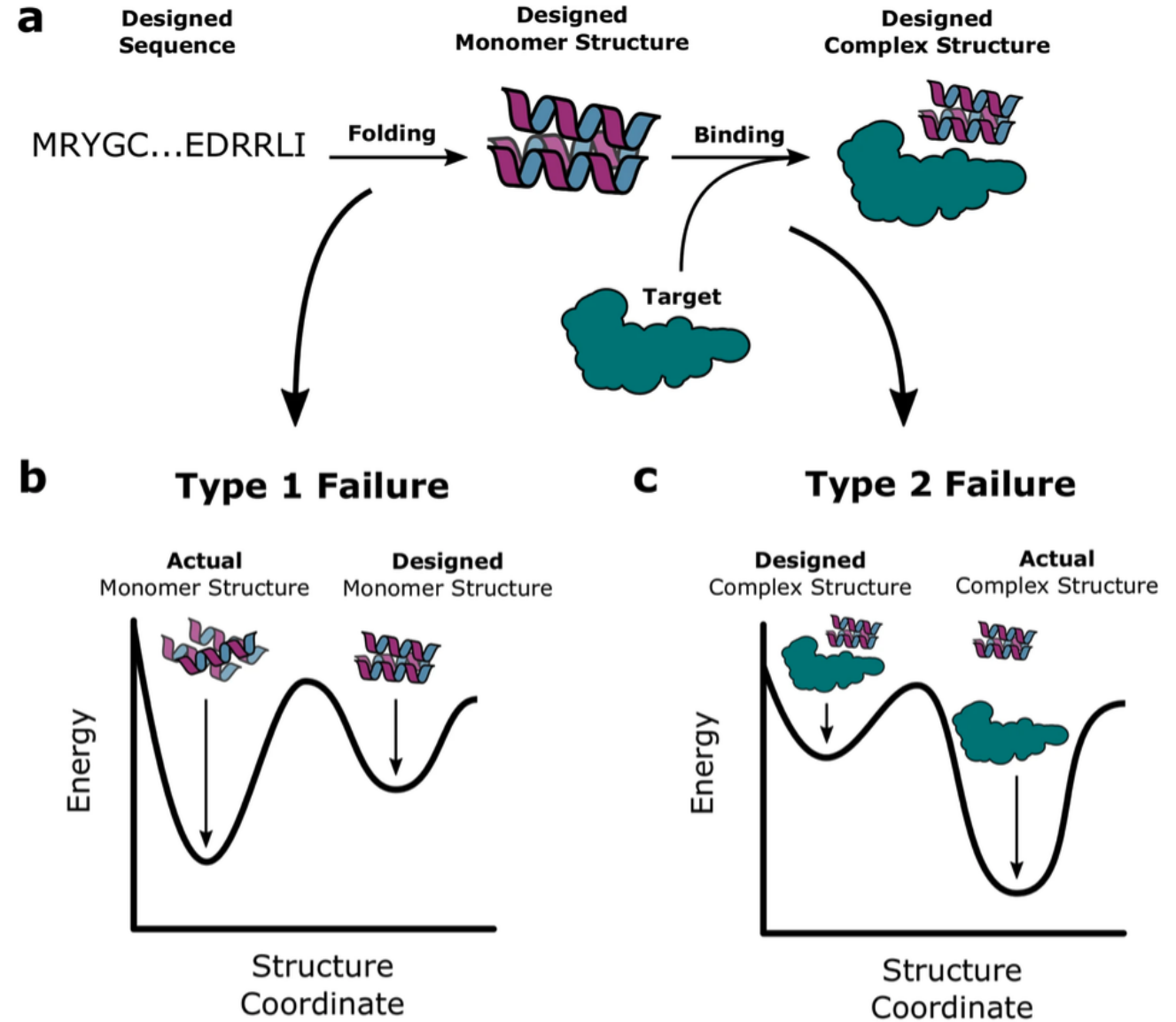
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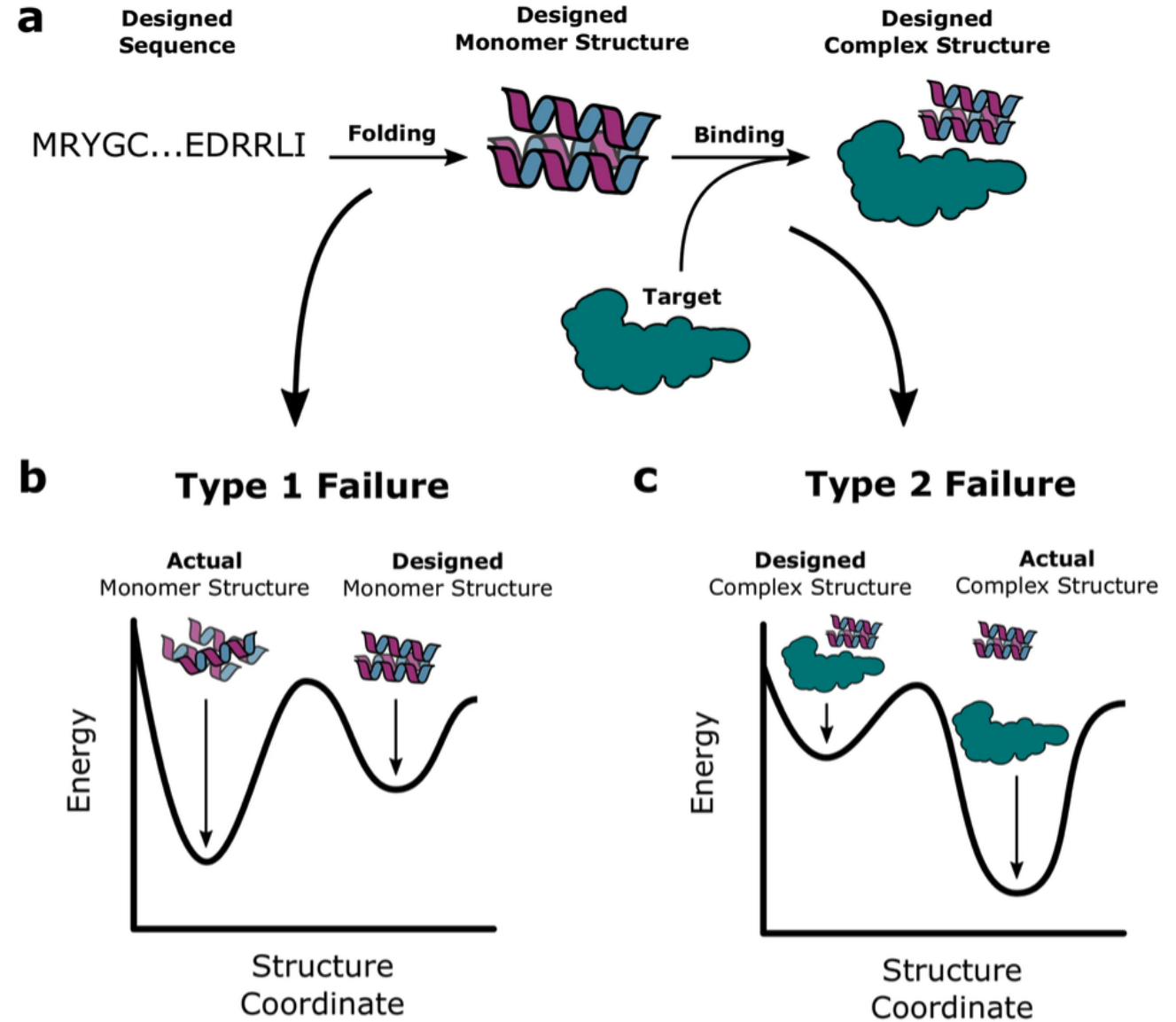
Type 1 failure

The binder does not fold correctly without the target.



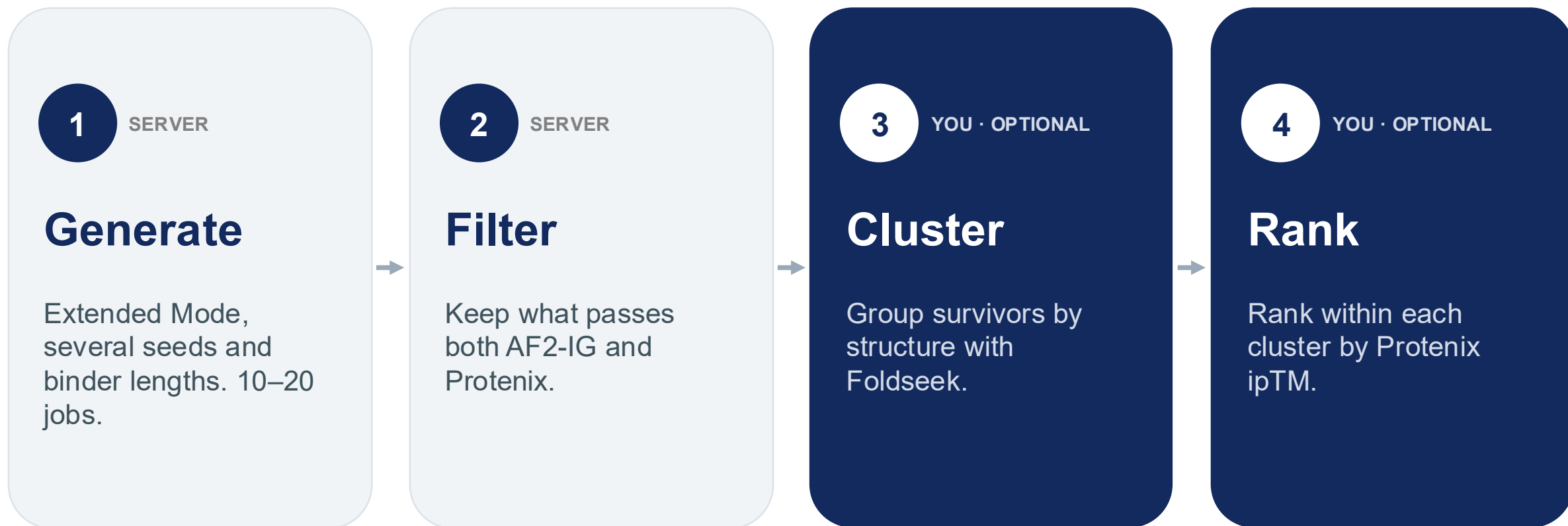
Type 2 failure

The binder folds correctly, but does not bind the target.

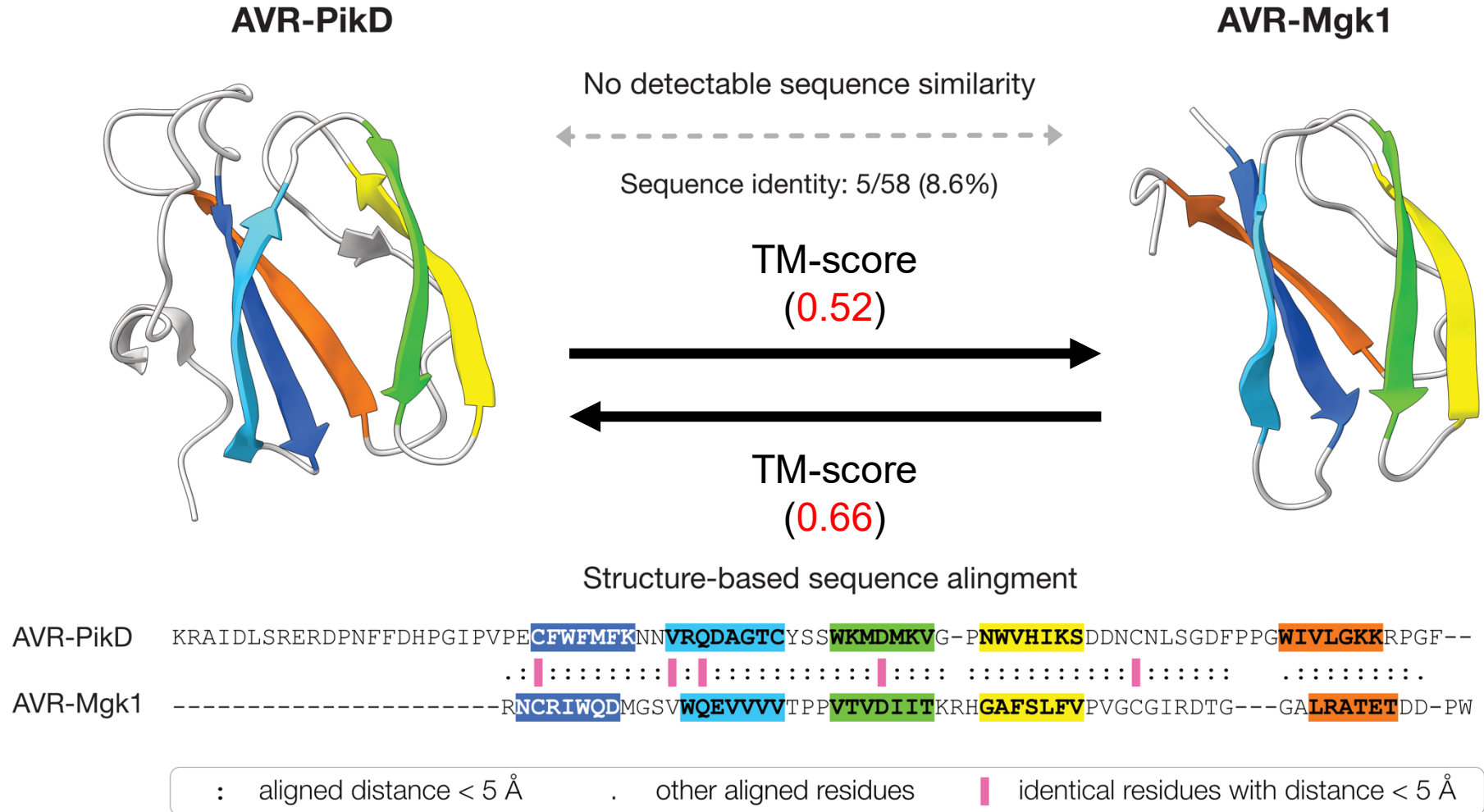


PXDesign's own recommendation for choosing designs

PXDesign's own advice, from protenix-server.com. Not a general rule.



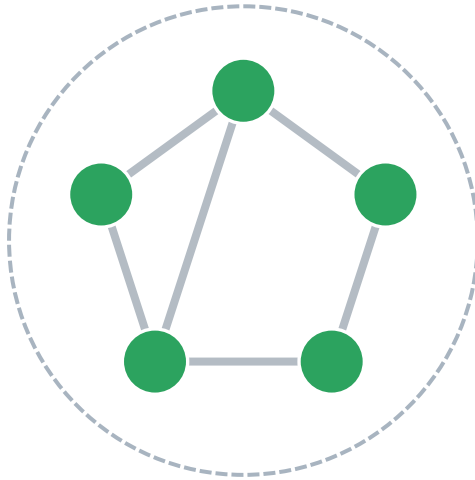
AVR-PikD and AVR-Mgk1 share the MAX fold despite having no sequence similarity



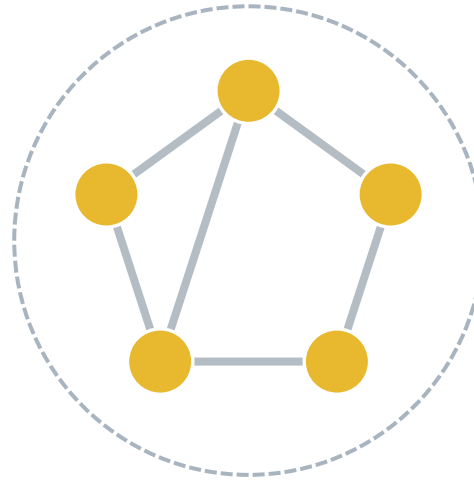
Sequence sorts real proteins into families

Every grey line is a BLAST hit.

family A



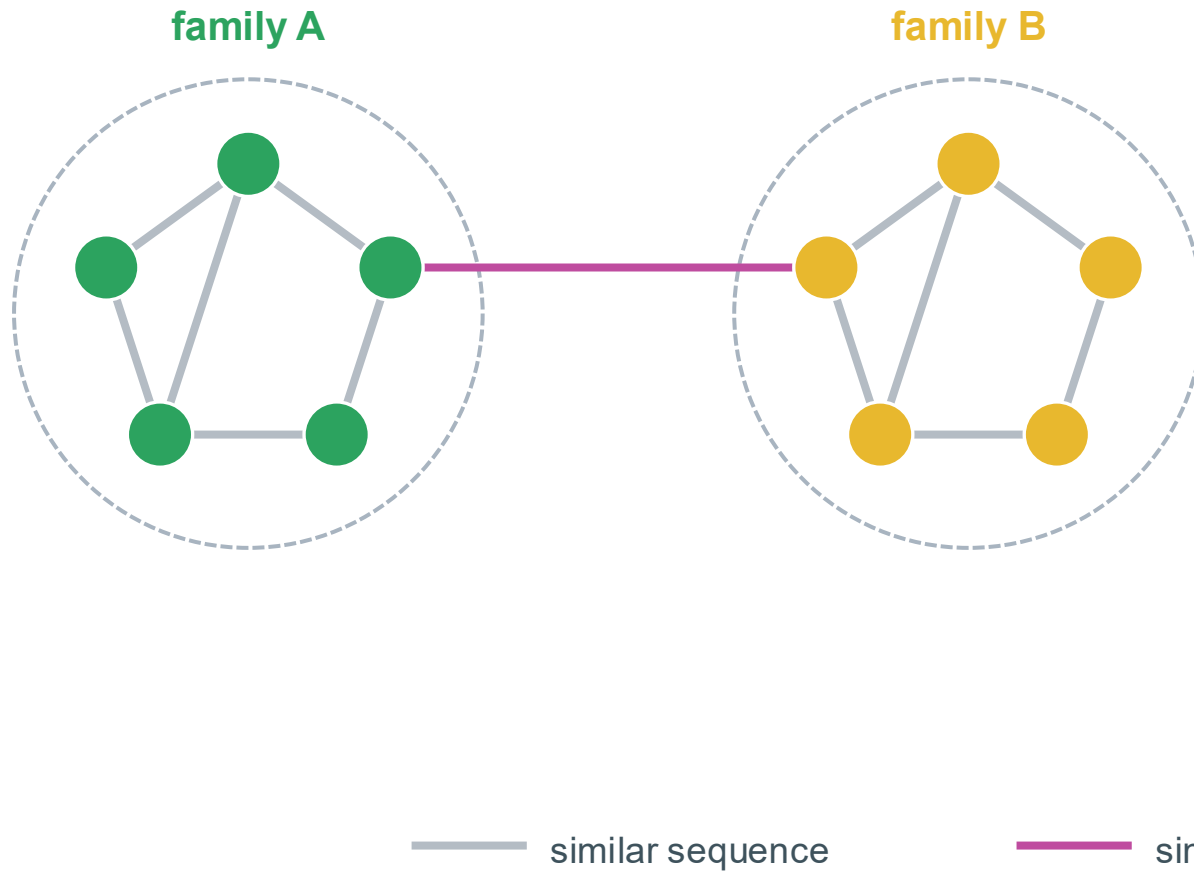
family B



— similar sequence

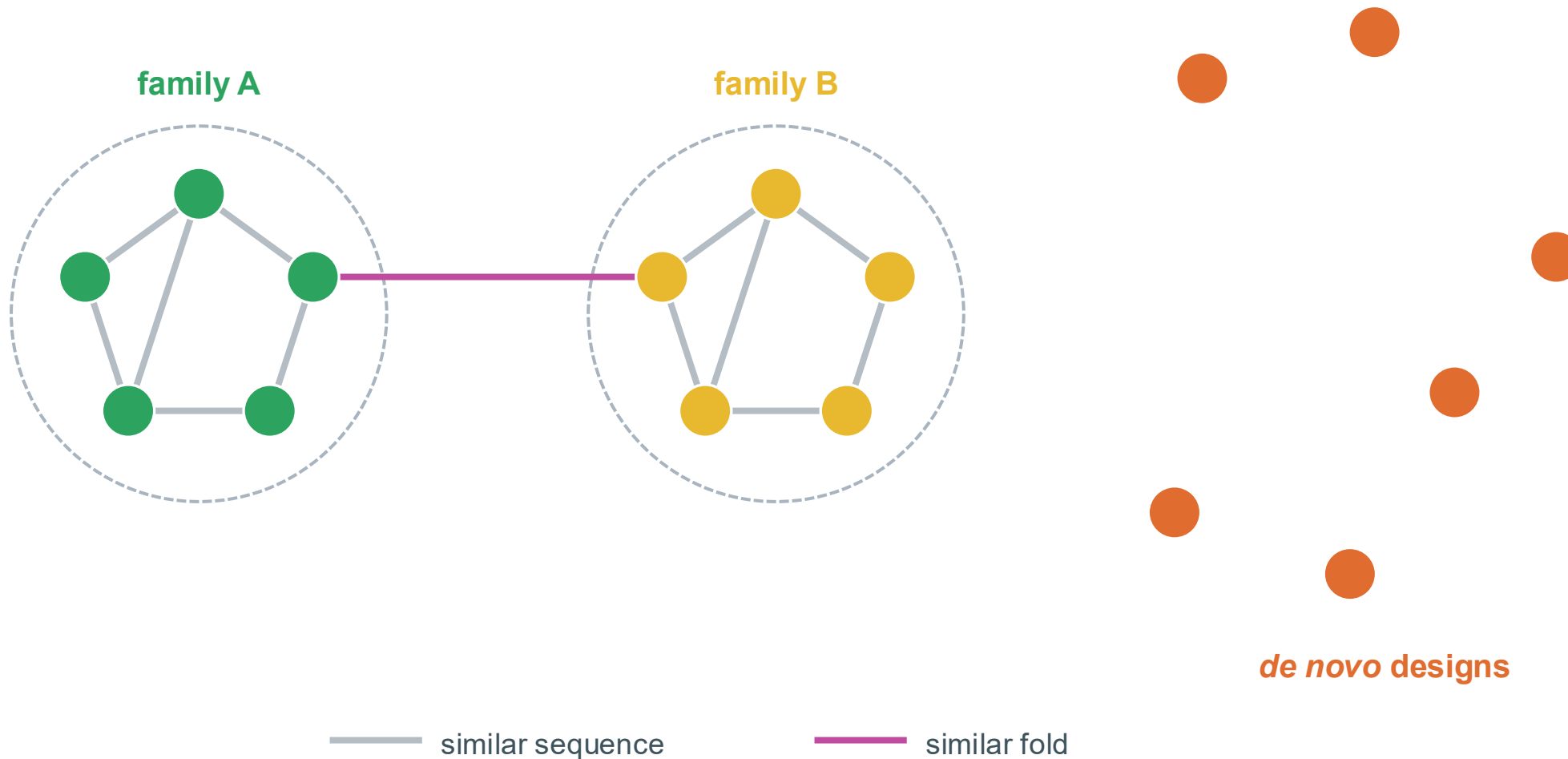
Structure links families with no shared sequence

One pink line, found by TM-align and Foldseek. Practical 3 ended here.



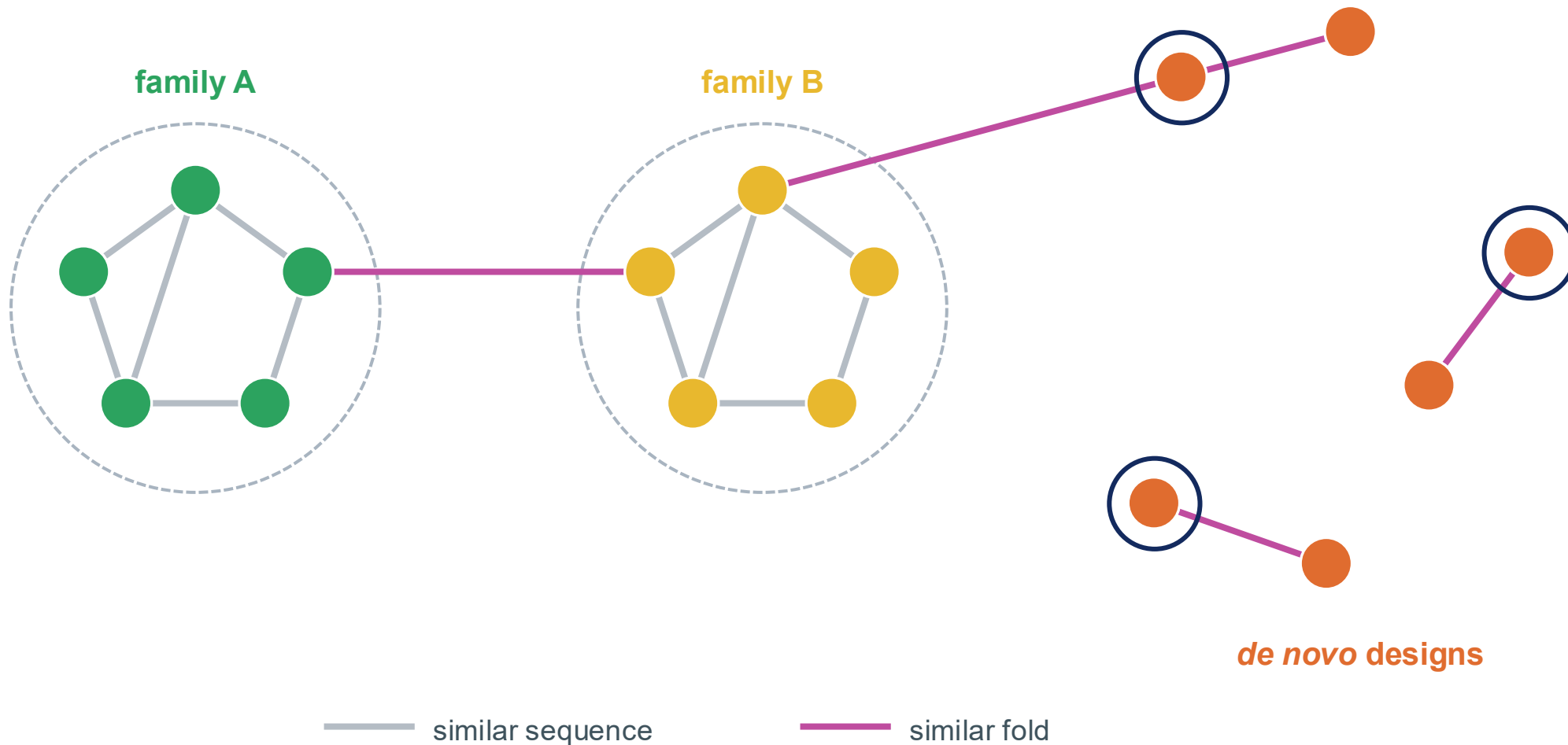
A designed binder has no relatives

No family, no homologues. A sequence search has nothing to find.



Structure is the only thing that connects them

Group the designs by fold, then rank inside each group.



Four things to take away

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- 1 **Generating designs is easy; getting a binder is not.**
- 2 **This field moves in weeks, not years.**
- 3 **No score yet tells you that a design will bind.**
- 4 **Much of what works is not in a journal.**

Programme

The same two-day programme runs at both venues (Espace Zmorda, then ISBS).

Day 1 Predict & Interpret

08:30–09:30	Registration
09:30–09:45	Welcome & Introductions
09:45–10:15	Introduction to AI for Genomics
10:15–11:00	Theoretical Introduction to AlphaFold
11:00–11:20	Break
11:20–12:30	 Practical 1: Predict
12:30–13:30	Lunch
13:30–15:00	<i>GOHREP it!</i> Interactive discussions
15:00–15:15	Break
15:15–17:00	 Practical 2: Interpret
17:00–17:15	Q&A

Day 2 Compare & Design

09:00–09:30	Day 2 Welcome & Registration
09:30–11:15	 Practical 3: Compare
11:15–11:30	Break
11:30–12:30	Model your own proteins
12:30–13:30	Lunch
13:30–14:45	 Practical 4: Design
14:45–15:05	Break
15:05–16:15	Genomics in an AI world
16:15–16:30	Summary
16:30–17:00	Q&A
17:00	Meeting close